

AR Formula Card

Every formula, conversion, and word-problem pattern for ASVAB Arithmetic Reasoning, in one place to keep next to you while you drill.

Order of operations (PEMDAS)

Parentheses, Exponents, Multiplication and Division (left to right), Addition and Subtraction (left to right). MD and AS share a tier — work left to right inside each.

P → E → M / D → A / S

Worked: $6 + 4 \times (3 + 2)^2 \div 5 = 6 + 4 \times 5^2 \div 5 = 6 + 4 \times 25 \div 5 = 6 + 100 \div 5 = 6 + 20 = 26$

Fraction → decimal → percent

Memorize this table. Half of AR is faster if you know that $3/8$ is .375 without thinking.

FRACTION	DECIMAL	PERCENT	FRACTION	DECIMAL	PERCENT
$1/2$	0.5	50%	$3/5$	0.6	60%
$1/3$	0.333	33.3%	$4/5$	0.8	80%
$2/3$	0.667	66.7%	$1/6$	0.167	16.7%
$1/4$	0.25	25%	$1/8$	0.125	12.5%
$3/4$	0.75	75%	$3/8$	0.375	37.5%
$1/5$	0.2	20%	$5/8$	0.625	62.5%
$2/5$	0.4	40%	$1/10$	0.1	10%

Distance / Rate / Time

D

R

T

D = R × T | R = D / T | T = D / R

Cover the variable you need. The other two stay in the visible position. Always check that minutes and hours match before you multiply.

Ex 1: A convoy travels 240 miles in 4 hours. Rate? $R = D / T = 240 / 4 = 60$ mph.

Ex 2: A jet flies at 450 mph for 2.5 hours. Distance? $D = 450 \times 2.5 = 1,125$ miles.

Work-rate (combined work)

$1/a + 1/b = 1/t$

Add individual rates as fractions, then flip the sum to get total time. a and b = solo times.

Ex: Painter A finishes in 4 hr, Painter B in 6 hr. Together? $1/4 + 1/6 = 3/12 + 2/12 = 5/12$ per hour. Time = $12/5 = 2.4$ hr.

Mixture / average

$$\text{Average} = \text{Sum} / \text{Count} \quad | \quad \text{Weighted avg} = (q_1v_1 + q_2v_2) / (q_1 + q_2)$$

Mixture trick: if you mix 3 gal at 20% with 2 gal at 50%, total acid = $0.20(3) + 0.50(2) = 0.6 + 1.0 = 1.6$ gal in 5 gal → **32%**.

Percent change & the “1.X” trick

$$\% \text{ change} = (\text{New} - \text{Old}) / \text{Old} \times 100$$

Increase by X%: multiply by 1.X. Up 15%? $\times 1.15$. **Decrease by X%:** multiply by $(1 - 0.X)$. Down 20%? $\times 0.80$.

Ex: A \$80 jacket goes up 25%. New price = $80 \times 1.25 = \text{\$100}$. Then 30% off = $100 \times 0.70 = \text{\$70}$.

Ratio / proportion (cross-multiply)

$$a / b = c / d \quad \rightarrow \quad a \times d = b \times c$$

Ex: 3 MREs cost \$12. 7 MREs cost? $3/12 = 7/x \rightarrow 3x = 84 \rightarrow x = \text{\$28}$.

Word-problem patterns ASVAB writers love

Recognize the cue, run the move. Don't solve the problem in your head — solve the pattern.

“How much faster / slower”

Think **rates**, not times. Convert both to mph (or units/hour) and subtract.

“Working together” / “at the same time”

Combined work-rate: $1/a + 1/b = 1/t$. Add the fractions, flip the sum.

“X% more” or “increased by X%”

Multiply the original by 1.X. Don't add then multiply again — it doubles up.

“What was the original price?”

Reverse percent. If sale price is 80% of original, divide sale price by 0.80.

“For every X, there are Y”

Set up a proportion. Cross-multiply. Don't reason it out in your head.

“On average” / “mean”

Sum / Count. For “what score do I need next time,” reverse it: Sum = Average $\times$ Count.

“Posts in a fence” / “floors in a building”

Off-by-one trap. A 100-ft fence with posts every 10 ft has **11 posts**, not 10.

“5 less than X” or “X fewer than Y”

Reverses order: 5 less than 12 = **12 – 5 = 7**, not $5 - 12$. Write it out before translating.

Unit conversions worth memorizing

LENGTH		VOLUME		TIME / WEIGHT	
1 ft	12 in	1 gal	4 qt	1 hr	60 min
1 yd	3 ft	1 qt	2 pt	1 min	60 sec
1 mi	5,280 ft	1 pt	2 cups	1 lb	16 oz

Test-day shortcuts.

- **Estimate first.** Round to clean numbers. If your estimate is 150 and one choice is 1,500, you can eliminate it without finishing the math.
- **Eliminate before solving.** Cross off impossible answers (negative when problem requires positive, fraction when answer must be whole).
- **Plug in answer choices.** When algebra setup is slow, drop choice B or C into the problem. If it works, you're done. If too high, try A. Too low, try D.
- **Re-read the question before selecting.** "What was the tax?" vs "What was the total?" The wrong answer is usually sitting in the choices.
- **Never leave a CAT question blank.** Unanswered questions at the end of the CAT trigger a scoring penalty. A guess beats a blank.

The 4-step word-problem method

1. **Read the whole problem.** The final sentence tells you what to find.
2. **Underline what's asked.** "How many more" $\neq$ "how many total."
3. **Map buzzwords to operations.** "Per" = divide. "Of" = multiply. "Is" = equals.
4. **Build the equation, then solve.** Don't calculate until the setup is on paper.

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